

IPL Sports Analytics & Win Probability Forecasting Project Report

Abstract

This project analyzes IPL (Indian Premier League) cricket match data to predict win probabilities and analyze player and team performance. It uses machine learning models, feature engineering, and interactive dashboards to deliver real-time win prediction, player segmentation, and venue intelligence across 16 IPL seasons.

Introduction

IPL matches generate large volumes of ball-by-ball data. Analyzing this data helps in understanding team strategies, player performance, and match outcomes. This project focuses on building a win probability engine and sports analytics platform to provide data-driven insights for cricket analysis.

Objectives

- Predict win probability during live IPL matches
- Analyze player batting and bowling performance
- Segment players into performance clusters
- Develop interactive dashboards for team and venue analysis
- Export data to Power BI for business intelligence reporting

Tools Used

- Python
- Scikit-learn and XGBoost
- Streamlit
- Power BI

Data Processing and Analysis

Steps performed:

- Data loading and cleaning from raw match and delivery files
- Feature engineering for live match context
- Model training and evaluation
- Dashboard development and Power BI export

Python Analysis

1. Objective:

Build a win probability prediction system using machine learning. The dataset is imported into Python using Visual Studio Code. Using the Pandas library, the dataset is loaded and inspected to understand its structure, columns, and data types.

2. Win Probability Prediction

Method Used:

- XGBoost Classifier (best model)
- Random Forest Classifier
- Logistic Regression (baseline)

Steps:

1. Load match and delivery datasets using Pandas
2. Select live match context features:
 - o Target score
 - o Runs needed
 - o Overs remaining
 - o Wickets remaining
 - o Current run rate
 - o Required run rate
3. Train XGBoost, Random Forest, and Logistic Regression models
4. Evaluate and compare model performance

Output:

- Win and loss probability for any match situation
- Model metrics including:
 - o XGBoost: 93.83% accuracy, ROC-AUC 0.9866
 - o Random Forest: 85.21% accuracy, ROC-AUC 0.9435
 - o Logistic Regression: 53.67% accuracy (baseline)

```
PS C:\Users\DELL\Desktop\SRAJAN_IPL_FORECASTING\scripts> python train_models.py
Training complete.
      model           purpose  accuracy  f1  roc_auc
  xgboost  live_match_context    0.9998 0.9998    1.0000
random_forest live_match_context    0.9948 0.9939    0.9999
logistic_regression match_level_baseline    0.4792 0.5614    0.5061
{
  "seasons": 17,
  "matches": 238,
  "deliveries": 57101,
  "best_model": "xgboost"
}
PS C:\Users\DELL\Desktop\SRAJAN_IPL_FORECASTING\scripts>
```

3. Player Segmentation

Objective:

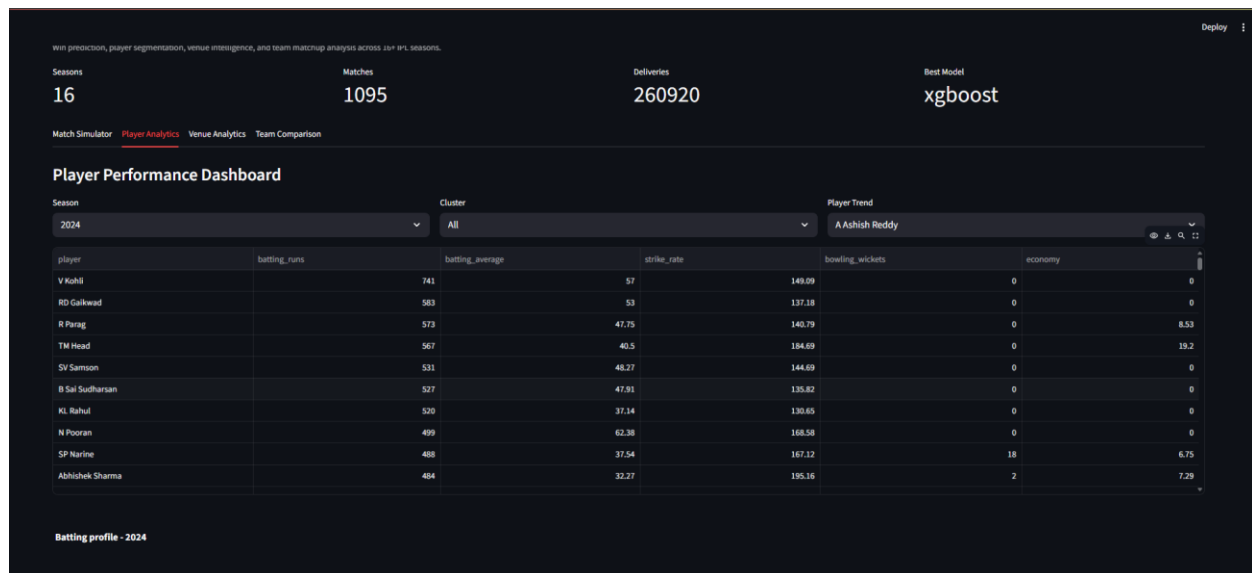
Segment IPL players into performance archetypes using clustering.

Steps:

5. Build player season features (batting and bowling stats)
6. Train K-Means clustering model
7. Assign cluster labels to each player

Output:

- Players categorized as:
 - Anchors
 - Power Hitters
 - Finishers
 - Economy Bowlers



Player performance dashboard

IPL Analytics Dashboard

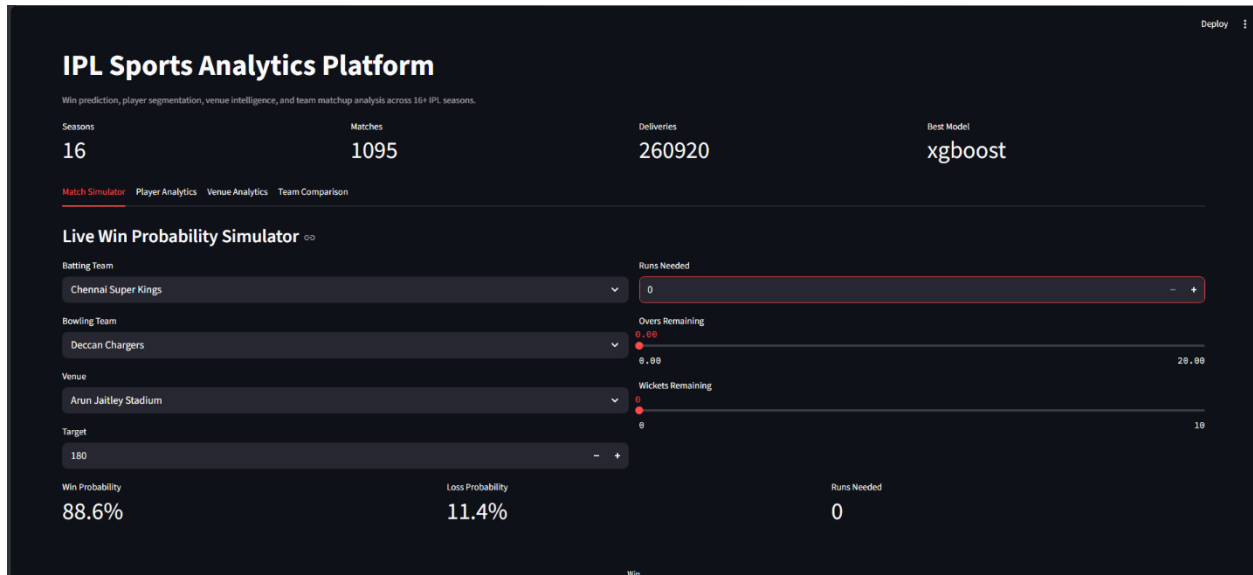
The processed dataset is loaded into a Streamlit application, where interactive visualizations are created to provide insights into match outcomes and player performance.

Key components of the dashboard include:

- Live Win Probability Simulator showing real-time win and loss percentages
- Model Comparison table covering XGBoost, Random Forest, and Logistic Regression
- Player Performance Dashboard for batting and bowling statistics
- Slicers/Filters for:

- o Season
- o Player cluster
- o Individual player

This dashboard helps users quickly simulate any match situation and monitor player performance across seasons.



IPL Analytics Dashboard

Venue and Team Analysis

The dashboard includes dedicated views for venue and team comparison.

Venue analysis includes:

- Average first innings score per venue
- Chase success rate at each venue
- Chase vs defend pattern heatmap

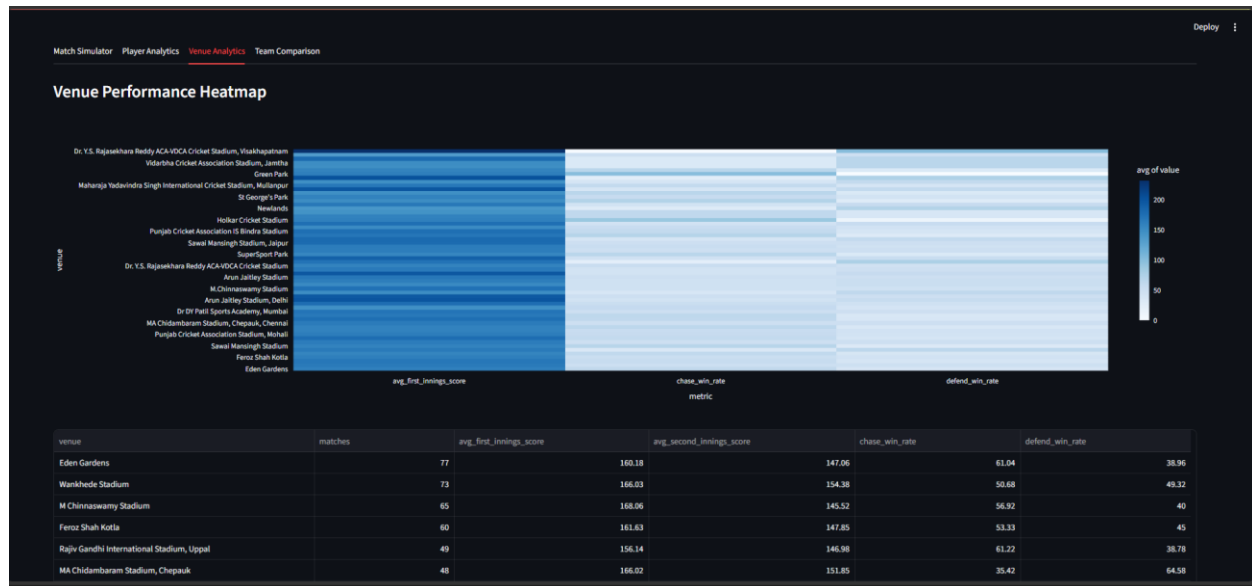
Team comparison includes:

- Head-to-head win and loss records

- Top performing players in each matchup
- Season-wise breakdown of encounters

This allows users to:

- Compare team strengths across different venues
- Identify patterns in match results
- Support data-driven team selection decisions



VENUE ANALYTICS

Project Outcome

This project shows how we can use data analysis and machine learning to understand IPL match data and predict match outcomes with high accuracy.

Instead of relying on manual observation, the system automatically analyses ball-by-ball data and provides win probabilities in real time.

Main Results:

- The system can predict match winner with 93.83% accuracy using XGBoost
- It helps analyse player performance easily across all seasons

- Dashboards make it easy to see and understand cricket data
- Decisions are based on data, not guesswork

Overall Benefit:

This helps coaches, analysts, and fans understand match dynamics, evaluate player form, and make informed decisions based on historical IPL data.